FOUR CHARACTERIZATIONS OF FUNCTORS MOD_A → MOD_B: TENSORING, RIGHT EXACT + DIRECT SUMS, ALL COLIMITS, RIGHT ADJOINT

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file functors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for functors.tex (fu-occupied.json, 2026-10-14); consistent with the duty manager's six-check dispatch.

Source reference. Original reference 09LN: sources/stacks-chow/tags.txt, source lines 12151–12151. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0038: sources/stacks-chow/tags.txt, source lines 125–125. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 010M: sources/stacks-chow/tags.txt, source lines 1332–1332. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let A and B be rings. Let $F : Mod_A \to Mod_B$ be a functor. The following are equivalent*

1. *F is isomorphic to the functor $M \mapsto M \otimes_A K$ for some $A \otimes_{\mathbf{Z}} B$-module K,*
2. *F is right exact and commutes with all direct sums,*
3. *F commutes with all colimits,*
4. *F has a right adjoint G.*

Proof. If (1), then (4) as a right adjoint for $M \mapsto M \otimes_A K$ is $N \mapsto \mathrm{Hom}_B(K, N)$, see Differential Graded Algebra, Lemma 09LN. If (4), then (3) by Categories, Lemma 0038. The implication (3) $\Rightarrow$ (2) is immediate from the definitions.

Assume (2). We will prove (1). By the discussion in Homology, Section 010M the functor F is additive. Hence F induces a ring map $A \to \mathrm{End}_B(F(M))$, $a \mapsto F(a \cdot \mathrm{id}_M)$ for every A-module M. We conclude that $F(M)$ is an $A \otimes_{\mathbf{Z}} B$-module functorially in M. Set $K = F(A)$. Define

$$M \otimes_A K = M \otimes_A F(A) \longrightarrow F(M), \quad m \otimes k \longmapsto F(\varphi_m)(k)$$

Here $\varphi_m : A \to M$ sends $a \to am$. The rule $(m, k) \mapsto F(\varphi_m)(k)$ is A-bilinear (and B-linear on the right) as required to obtain the displayed $A \otimes_{\mathbf{Z}} B$-linear map.

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This construction is functorial in M, hence defines a transformation of functors $- \otimes_A K \to F(-)$ which is an isomorphism when evaluated on A. For every A-module M we can choose an exact sequence

$$\bigoplus\nolimits_{j \in J} A \to \bigoplus\nolimits_{i \in I} A \to M \to 0$$

Using the maps constructed above we find a commutative diagram

$$\begin{array}{ccccccc} (\bigoplus_{j \in J} A) \otimes_A K & \longrightarrow & (\bigoplus_{i \in I} A) \otimes_A K & \longrightarrow & M \otimes_A K & \longrightarrow & 0 \\ \downarrow & & \downarrow & & \downarrow & & \\ F(\bigoplus_{j \in J} A) & \longrightarrow & F(\bigoplus_{i \in I} A) & \longrightarrow & F(M) & \longrightarrow & 0 \end{array}$$

The lower row is exact as F is right exact. The upper row is exact as tensor product with K is right exact. Since F commutes with direct sums the left two vertical arrows are bijections. Hence we conclude. $\square$